Experiment 6

Student Name: Shashwat **UID: 22bcs117 43**
Branch: CSE **Section/Group: 641A**
Semester: 6 **Date of Performance:**
Subject Name: AP LAB-II **Subject Code: 22CSP-351**

1. Aim:

a. To find and implement the maximum depth of Binary Tree.

b. To develop an algorithm for Binary Tree Inorder traversal.

2. Objective:

To implement and analyze maximum depth of Binary Tree.

To develop an algorithm for Binary Tree Inorder traversal.

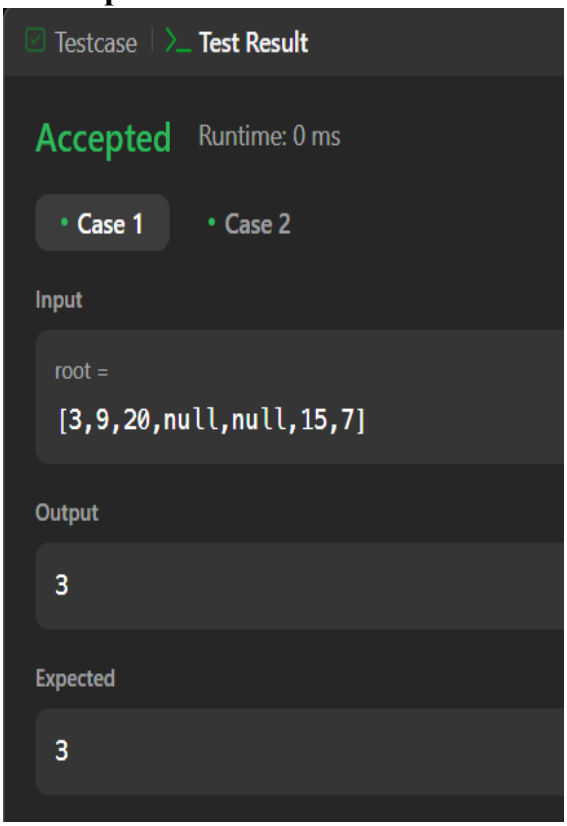
3. Implementation/Code:

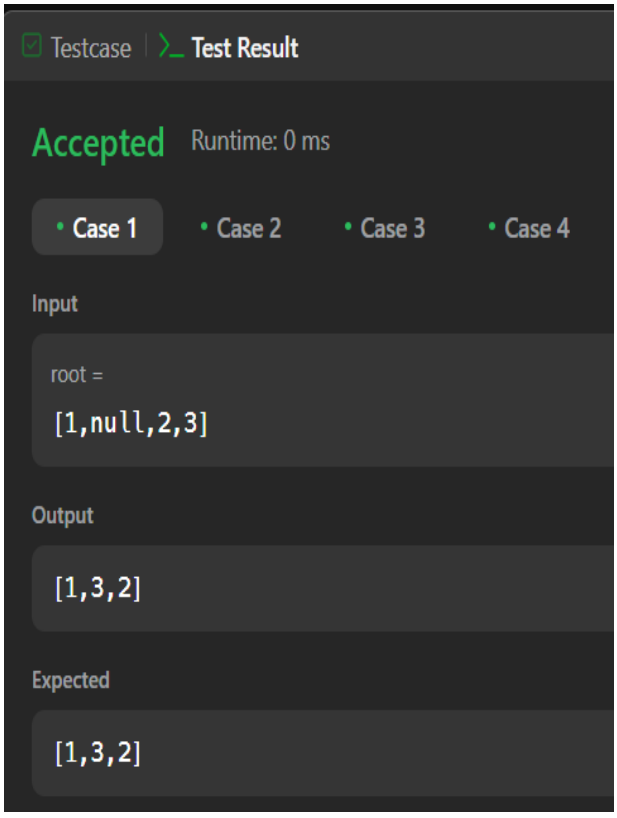
a.
```
class Solution {
public:
    int maxDepth(TreeNode* root) {
        if (!root) return 0;
        int leftDepth = maxDepth(root->left);
        int rightDepth = maxDepth(root->right);
        return 1 + max(leftDepth, rightDepth);
    }
};
```

b.
```
class Solution {
public:
    vector<int> inorderTraversal(TreeNode* root) {
        vector<int>ans;
        in(root,ans);
        return ans;
    }
    void in(TreeNode* root,vector<int>&ans)
    {
        if(root==NULL)
```

```
        return;
        in(root->left,ans);
        ans.push_back(root->val);
        in(root->right,ans);
    }
};
```

4. Output:





5. Learning Outcome:

- Understand string manipulation techniques in C++.
- Implement efficient algorithms for detecting cyclic rotations.
- Apply mathematical approaches to solve missing number problems.
- Utilize standard library functions like accumulate and find.
- Enhance problem-solving skills through algorithm design and analysis.